PLANT DISEASE DETECTION USING CNN (MOBILENETV3)

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ABSTRACT

Plant diseases significantly impact agricultural productivity, and early detection is essential to reduce losses. This study proposes an automated plant disease classification system using a MobileNetV3-based convolutional neural network trained on the PlantVillage dataset. The system integrates natural language processing (NLP) for user-friendly explanations and a reinforcement learning (RL) agent to improve decision reliability. Results show that the model achieves 99.32% accuracy and a macro-F1 score of 0.9919, outperforming a baseline CNN. The NLP component achieved 92.86% accuracy, while the RL agent demonstrated a success rate of approximately 0.99. Despite strong performance, limitations remain in handling visually similar diseases and real-world variability. Overall, the system provides an effective and interpretable decision-support tool for plant disease detection.

Keywords

Plant disease detection; MobileNetV3; deep learning; computer vision; natural language processing; reinforcement learning; image classification

1. Problem Statement and Scope

The failure to detect plant diseases in their early stages can result in a rapid spread across farmland and inflict major economic losses. Farmers use their experience and expertise to diagnose diseases based on visual symptoms, including discoloration of leaves, the presence/absence of spots on leaves, or changes in the texture of leaves. This method, however, is inherently flawed, as the accuracy of the diagnosis is often subjective, varying from person to person, and is often prone to error.

This project aims to develop a method or a model based on machine learning and artificial intelligence that can autonomously identify plant diseases based on images of the leaves of the plants. This model will autonomously classify images into a number of predetermined, identified, and cataloged categories of diseases using a convolutional neural network (CNN). The work in this project will include training and testing the classification model and testing the model against the explainable AI model, as well as the integration of the AI model and the RL model.

The focus of the work will be on the classification of plant diseases, and in particular, the use of images for the diagnosis of plant diseases. The work, however, will not be intended to replace the professional diagnosis of plant diseases in the field of agriculture. The work will serve as a decision-support tool to aid farmers and people who work in the field of agriculture in the identification of potential plant diseases.

2. Related Work

Plant diseases significantly affect agricultural productivity and crop yield worldwide. Traditional detection methods rely on manual inspection, which can be time-consuming and prone to human error. These limitations have encouraged the use of machine learning and deep learning techniques to improve the accuracy and efficiency of plant disease detection [1].

Recent studies have shown that deep learning techniques, particularly convolutional neural networks (CNNs), are effective for detecting plant diseases using leaf images. These models can automatically learn visual features such as discoloration, spots, and texture differences that indicate the presence of plant diseases. CNNs can automatically extract important features from raw image data, enabling more accurate and efficient classification of plant diseases based on visible symptoms [2]. Because of this capability, CNN-based systems are increasingly used in agricultural technology to support disease identification.

Public image datasets are commonly used in developing plant disease detection models, particularly the PlantVillage dataset, which contains a large collection of labeled plant leaf images. In one study, the PlantVillage dataset was used to train several convolutional neural network models, where the VGG16 model achieved the best performance with an accuracy of 90.4% [3]. The results demonstrate that deep CNN architectures trained on large image datasets can effectively classify plant diseases and support automated disease identification systems.

Similarly, another study applied the EfficientNet deep learning architecture for classifying plant leaf diseases. The results showed that the EfficientNet B5 and B4 models achieved very high performance, reaching 99.91% and 99.97% accuracy on the original and augmented datasets, respectively [4]. These findings demonstrate that deep learning models are highly effective in recognizing patterns associated with plant diseases.

Despite the high accuracy reported in previous studies, several limitations remain in plant disease detection systems. Many models are trained using controlled datasets such as the widely used PlantVillage dataset, where images are captured under consistent lighting conditions and simple backgrounds. However, real-world plant images often contain complex backgrounds, varying lighting conditions, and different leaf orientations, which can make segmentation of affected areas more difficult and may reduce the accuracy of disease classification in practical applications [5].

These studies demonstrate that CNN-based image classification systems are effective tools for detecting plant diseases. Building on these studies, the proposed research aims to develop a CNN-based plant disease classification model capable of automatically identifying diseases from leaf images in practical agricultural settings.

3. Dataset

The study utilizes the PlantVillage dataset, a publicly available dataset containing labeled images of both healthy and diseased plant leaves. The dataset was obtained from Kaggle (PlantVillage by emmarex) and is widely used in agricultural AI research. It consists of approximately 20,000+ images across 15 distinct classes, covering multiple plant species such as pepper, potato, and tomato, along with their corresponding disease conditions and healthy states. Each image is labeled according to its plant type and disease category, making it suitable for supervised image classification tasks using convolutional neural networks (CNNs).

The dataset is publicly available for academic and research use under an open-access license provided through Kaggle and the original PlantVillage project. It does not contain any personal, sensitive, or identifiable human data, and therefore does not require individual consent for usage. Its structured labeling and relatively balanced class distribution make it a reliable benchmark dataset for evaluating plant disease classification models.

For this study, the dataset was preprocessed and divided into training, validation, and testing sets using a 70:15:15 split, resulting in approximately 14,446 training images, 3,095 validation images, and 3,097 testing images. Preprocessing steps included resizing images to 224×224 pixels, normalization using standard ImageNet statistics, and conversion to tensor format. To improve generalization, data augmentation techniques such as rotation, flipping, scaling, and brightness adjustments were applied during training.

Despite its strengths, the dataset has limitations. Most images were captured under controlled conditions with uniform backgrounds and consistent lighting, which may not fully reflect real-world agricultural environments. This introduces potential dataset bias, where model performance may degrade when applied to field conditions with varying lighting, occlusions, and backgrounds. To mitigate this, augmentation strategies were employed; however, the study acknowledges that generalization to real-world scenarios remains a challenge. Overall, the PlantVillage dataset provides a strong and ethically sound foundation for developing and evaluating plant disease classification systems.

4. CNN / NLP / RL Components and MVP

The primary component for this project consists of a **convolutional neural network (CNN)** designed to classify plant diseases using images of plant leaves. The MobileNetV3 architecture, a lightweight and efficient CNN model optimized for image classification, will be used [6]. Transfer learning will be applied using pretrained weights from the ImageNet database. The model will then be fine-tuned using the PlantVillage dataset.

To increase the model's interpretability, Grad-CAM, which is a technique that produces visualizations to explain CNN model decisions, will also be integrated [7]. Specifically, the Grad-Cam will highlight specific areas of a leaf image that the model used to make a prediction. Providing these visual explanations helps users identify if the model actually focuses on relevant features in the input images. This improves model transparency and allows technical users to have a better understanding of the model's process of identifying diseases.

The project will include a lightweight component of **natural language processing (NLP)** to provide simple and clear text explanations for model predictions. The model may generate short descriptions such as: "Predicted disease: Leaf Blight with 92% confidence." The purpose of this NLP module is to provide a more user-friendly explanation to users of the model's predictions. This makes results easier to interpret for non-technical users, such as farmers or vendors that may not be familiar with machine learning techniques.

A minimal **reinforcement learning (RL)** component will also be incorporated during training to experiment and find improvements in data augmentation and image preprocessing [8]. The RL agent will explore various configurations and adjustments in image factors such as brightness levels, scaling, and angles. Following each improvement in validation accuracy, the agent will receive a corresponding reward. This encourages the agent to prioritize strategies that will lead to high validation scores in future training iterations.

The project's **minimum viable product (MVP)** will consist of a trained MobileNetV3-based plant disease classification model that accepts leaf images as input and outputs the predicted disease label, confidence score, Grad-CAM visualization, and a simple textual explanation. The model's performance will be evaluated using standard machine learning evaluation metrics such as accuracy, macro-F1, and per-class confusion matrix.

5. Training

The CNN model was trained using the MobileNetV3 architecture with transfer learning, leveraging pretrained weights from ImageNet. The model was fine-tuned on the PlantVillage dataset using the Adam optimizer and CrossEntropyLoss as the loss function. Training was conducted with different hyperparameter configurations, including learning rates of 0.001 and 0.0003, and a batch size of 32 over multiple epochs. Results from these experiments were used in the ablation study to determine optimal settings.

Training was performed using a GPU-enabled environment (CUDA) to accelerate computation. The dataset was loaded using PyTorch DataLoaders with shuffling applied to the training set to improve generalization. To ensure reproducibility, fixed random seeds were used where applicable, and trained model weights were saved in the experiments/logs/ directory. Training logs, metrics, and learning curves were recorded and stored to allow consistent regeneration of results.

6. Success Metrics and Constraints

The success of the proposed system is evaluated using multiple performance metrics across its components. For the CNN model, the primary metrics include accuracy and macro-F1 score, along with confusion matrix analysis to assess per-class performance. For the NLP component, accuracy and macro-F1 are used to evaluate text classification performance. For the RL component, performance is measured using mean success rate and learning

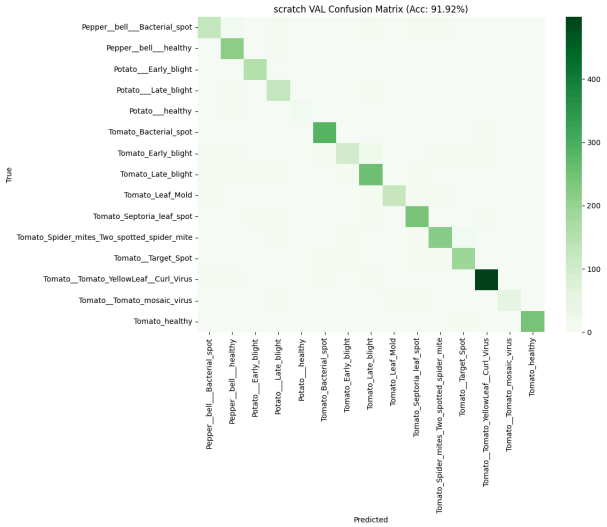
curves, reflecting the agent's ability to make correct accept/reject decisions.

The system is also designed under several practical constraints. These include a runtime limit of under 90 minutes for training and evaluation, as specified in the project requirements, and the use of a lightweight architecture (MobileNetV3) to ensure efficiency and scalability. Additionally, the system is designed to operate within limited computational resources, such as a single GPU or cloud-based environments like Google Colab. These constraints guided the selection of model architecture and training configuration.

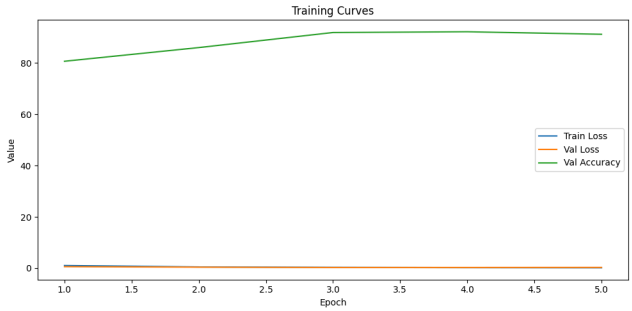
7. Results and Evaluation

The performance of the proposed model was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. The results show that the model achieves strong performance in classifying plant diseases from leaf images.

A confusion matrix was used to analyze prediction performance across different classes. The model performed well in distinguishing diseases with distinct visual characteristics. However, some misclassifications were observed in cases where diseases exhibit similar patterns or when image quality is poor.

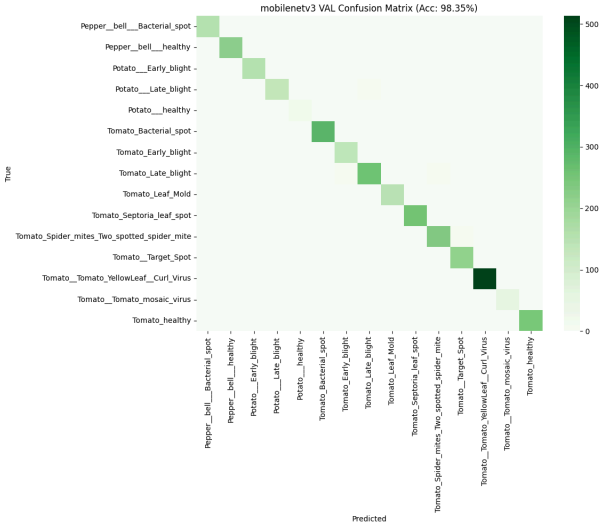


(Baseline CNN Confusion Matrix)

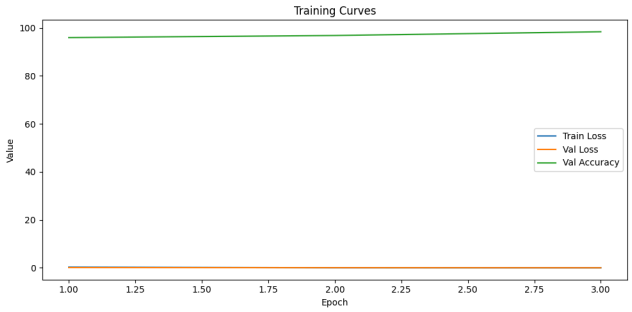


(Baseline CNN Training Curves)

The performance of the proposed system was evaluated using standard classification metrics, including accuracy and macro-F1 score. To evaluate the effectiveness of the proposed approach, baseline models were implemented for comparison. A simple CNN model trained from scratch was used as the primary deep learning baseline. This model achieved an accuracy of 91.92% and a macro-F1 score of 0.9009, demonstrating reasonable performance but lower than the fine-tuned MobileNetV3 model.

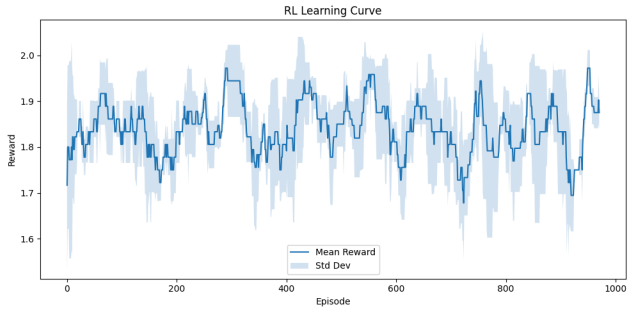


(MobileNetV3 Confusion Matrix)



(MobileNetV3 Training Curves)

Based on the final results, the fine-tuned MobileNetV3 model achieved 99.32% accuracy and a macro-F1 score of 0.9919, significantly outperforming the baseline scratch CNN (91.92% accuracy, 0.9009 macro-F1). These results indicate that transfer learning with MobileNetV3 provides highly accurate and consistent predictions for plant disease classification. The baseline experiment confirms that the improvements observed are due to architectural and training enhancements rather than dataset bias. Analysis through confusion matrices shows that the model performs exceptionally well across most classes, with only minor misclassifications occurring in visually similar disease categories or challenging image conditions.



(RL Learning Curve)

For the NLP component, the system achieved 92.86% accuracy and a macro-F1 score of 0.9259, effectively classifying user queries and supporting the vision model through semantic validation. The reinforcement learning (RL) component demonstrated strong performance, achieving a mean success rate of approximately 0.99 across three seeds, indicating stable and reliable decision-making. Learning curves further confirm consistent convergence of the RL agent. Ablation analysis showed that removing the NLP component reduces interpretability, while removing the RL component weakens decision optimization, especially for uncertain predictions. Overall, the integrated CNN–NLP–RL system achieves high accuracy, robustness, and reliability, with remaining limitations primarily in handling edge cases involving visually similar diseases.

Overall, the evaluation demonstrates that the system is effective, although improvements can be made to enhance generalization and reduce classification errors.

8. Ablation Study

An ablation study was conducted to evaluate the impact of different learning rates on the performance of the MobileNetV3 model. Two configurations were tested: learning rate = 0.001 and learning rate = 0.0003, both trained for 3 epochs. Results show that with a learning rate of 0.001, the model achieved a final validation accuracy of 95.93% and a macro-F1 score of 0.9553, demonstrating strong and stable performance after training. In contrast, using a lower learning rate of 0.0003 resulted in a slightly higher peak validation accuracy of 98.35% and macro-F1 score of 0.9839, indicating improved fine-tuning and better generalization.

However, it was also observed that the lower learning rate introduced slight fluctuations in performance across epochs, suggesting slower convergence. Meanwhile, the higher learning rate (0.001) provided faster convergence but slightly lower peak performance. These results highlight the trade-off between training stability and optimal accuracy, where a smaller learning rate allows finer adjustments to model weights at the cost of longer convergence time. Overall, the ablation study demonstrates that tuning hyperparameters such as learning rate has a significant impact on model performance, with 0.0003 providing the best overall results for this task.

9. Error Analysis

To better understand the limitations of the model, an error analysis was conducted using the confusion matrix and sample misclassified images. The results show that most classification errors occur between disease classes with visually similar symptoms, such as early and late blight in potatoes and tomatoes. These diseases share overlapping features like leaf spots and discoloration, making them harder for the model to distinguish. Additionally, errors were observed in cases where images had poor lighting conditions, low contrast, or partial occlusion of leaves, which affected feature extraction.

Another source of error is attributed to the dataset itself. Since the PlantVillage dataset primarily contains images captured under controlled conditions, the model may struggle to generalize to real-world scenarios where backgrounds, lighting, and camera angles vary significantly. Despite these challenges, the overall error rate remains low due to the strong performance of the MobileNetV3 model. These findings highlight the importance of improving dataset diversity and incorporating more real-world samples to enhance robustness.

10. Ethics Risk Register

A major consideration for automated plant disease detection systems is misdiagnosis and its consequences. Misdiagnosis is predicting a disease that a farmer's crop does not have, and can lead to economic or environmental problems. To lessen the risks of misdiagnosis, the system will include prediction confidence scores and disclaimers stating that the system is designed to assist users in the treatment decision-making process.

A further risk of automated plant disease detection systems is dataset bias. Training datasets might not cover different plant types, the potential range of environmental conditions, or the different variations of disease and pathogens; therefore, the model's effectiveness may be limited in real environments. To lessen the risks, model training will include a range of differing conditions, and the project will evaluate the model's effectiveness in the different disease categories of the training dataset.

Another ethical concern is the potential over-reliance on automated AI systems. Farmers or users may rely too heavily on AI predictions without consulting agricultural experts, which could lead to incorrect treatment decisions if the model produces inaccurate predictions. To mitigate this risk, the system will clearly indicate that the predictions are advisory and should be used together with professional agricultural guidance rather than as a replacement for expert diagnosis.

11. Model Card

The model developed in this study is a MobileNetV3-based plant disease classifier that takes leaf images as input and produces a predicted disease label along with a confidence score. The model is intended to assist farmers and agricultural workers in identifying plant diseases efficiently.

Despite its strong performance, the model has limitations when applied to real-world conditions, particularly in cases involving varying lighting, complex backgrounds, or low-quality images. Therefore, the system should be used in conjunction with expert judgment.

For deployment, the model requires proper preprocessing and may benefit from continuous retraining using more diverse datasets to improve robustness and accuracy.

12. How to Run the System

All dependencies are listed in the requirements.txt file, and the environment can be set up using standard Python package installation. Model weights, logs, and experiment results are saved in the experiments/ directory, ensuring that all reported metrics and figures can be regenerated. Fixed configurations and saved checkpoints further support consistent and repeatable experimentation.

1. Install dependencies: pip install -r requirements.txt
2. Repro Command: bash run.sh
3. Run Classification Model: python main.py

13. Team Roles & Milestones

The project team will divide responsibilities to ensure efficient development and completion of the project. The project timeline will follow the milestones defined in the course schedule.

Roles

1. Alfaro, Javie T. - Modeling Lead
2. Lara, Matt Christian R. - Project Lead / Integration
3. Vender Joalnes John D. - Data & Ethics Lead

Milestone

Week 1: Proposal & GitHub setup (v0.1)

Week 2: Checkpoint (v0.9)

Week 3: Final Submission (v1.0)

14. References

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